Harder Geometrical Problems — problem 12.3.2.72

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Consider the following theorems :
(U) : *Circumscribed quadrilateral* $\iff \alpha + \gamma = \beta + \delta = 180°$.
(I) : *Inscribed quadrilateral* $\iff a + c = b + d$.
(A) : *Area of quadrilateral* $A = \sqrt{abcd}$.
Prove that $(U), (I) \Rightarrow (A), \quad (U), (A) \Rightarrow (I), \quad (I), (A) \Rightarrow (U)$.